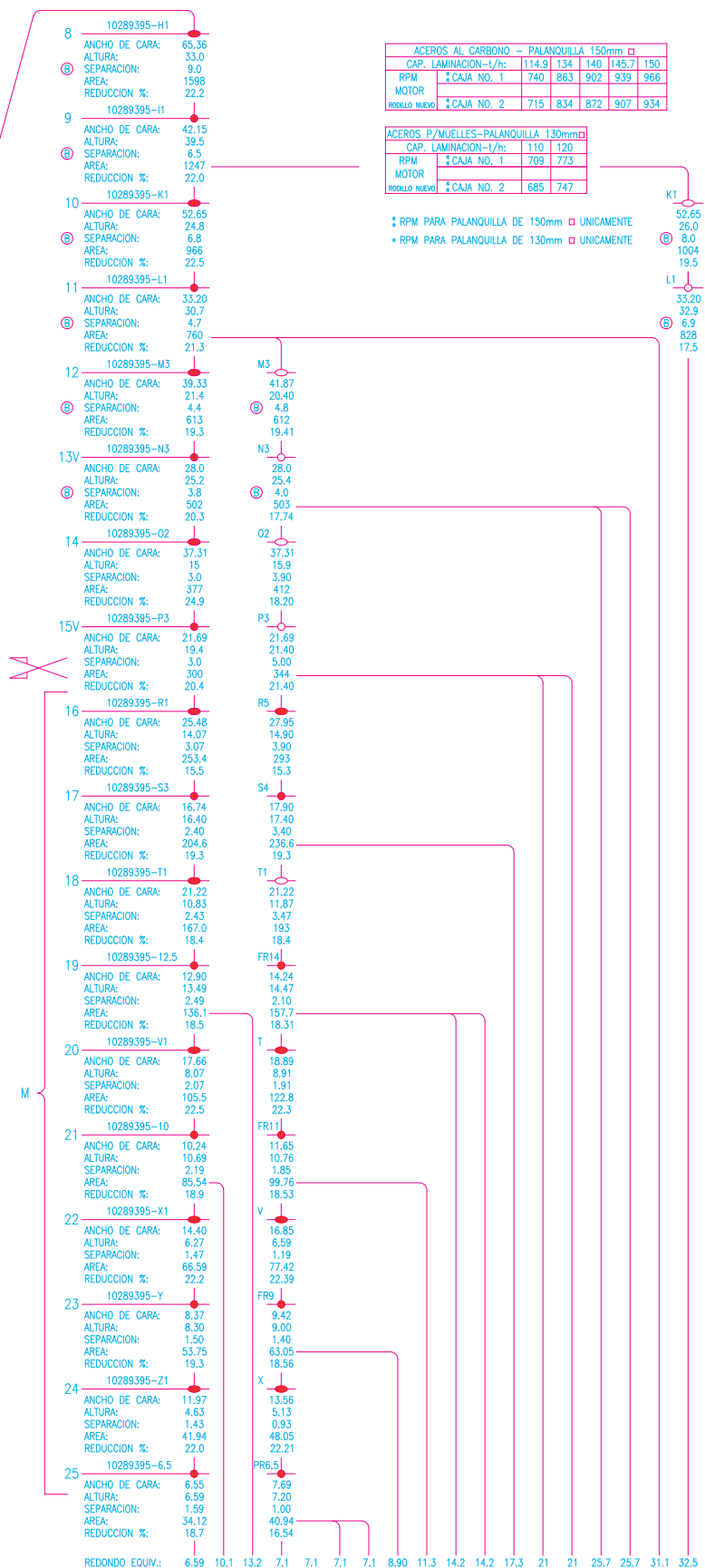


Category	Value
1	52.65
2	26.0
3	8.0
4	1004
5	19.5
6	33.20
7	32.9
8	6.9
9	828
10	17.5



REFERENCIA MORGAN: 10289394

[illegible]

		1	2	3	4	5	6	7	8	9	10	11	12
A													
B													
C													
D													
E													
F													
G													
H													

FEED DIAMETER:
FEED AREA:
ENTRY M/S:

11.3
99.6
27.6

13.2
136.1
20.2

14.2
158.6
17.5

16.1
204.6
13.6

17.3
235.6
11.8

19.5
300.0
9.26

21.0
346.0
7.99

22.0
380.0
7.28

25.7
520.0
5.33

29.0
661.0
4.19

31.1
760.0
3.64

32.5
828.0
3.34

PASS:
WIDTH ON FACE:
HEIGHT:
PARTING:
AREA:
REDUCTION %:

OV_9.0RM
16.72
6.80
1.00
78.50
21.1

OV_9.5RM
15.22
8.02
1.00
84.88
14.7

OV_10.0RM
19.14
7.32
1.20
99.75
26.7

OV_10.5RM
19.19
8.14
1.20
107.85
20.7

OV_11.0RM
19.68
8.56
1.20
120.00
24.3

OV_11.5RM
18.27
9.64
1.20
126.85
20.0

OV_12.0RM
18.91
10.42
1.20
136.14
14.2

OV_12.5RM
22.35
9.92
1.20
155.96
23.8

OV_13.0RM
23.11
10.64
1.20
166.27
18.7

OV_13.5RM
25.16
10.72
1.20
183.39
22.2

OV_14.0RM
24.16
11.72
1.20
192.51
18.3

OV_14.5RM
21.44
12.98
1.20
198.55
15.7

OV_15.0RM
28.22
11.96
1.40
230.06
23.2

OV_15.5RM
27.25
12.92
1.40
240.11
20.0

OV_16.0RM
29.31
12.94
1.40
262.06
24.3

OV_17.0RM
27.27
14.81
1.75
283.16
18.2

OV_18.0RM
27.46
16.10
1.40
313.11
17.6

OV_19.0RM
32.66
15.92
1.40
371.45
28.6

OV_20.0RM
33.26
17.20
2.00
404.02
22.3

OV_21.0RM
34.51
18.70
2.00
435.41
16.3

OV_22.0RM
37.96
18.86
1.40
498.31
24.6

OV_23.0RM
39.13
20.28
1.40
534.00
19.2

OV_24.0RM
39.25
21.12
2.10
586.24
22.9

OV_25.0RM
40.56
22.16
1.40
633.27
23.5

OV_26.0RM
41.49
23.50
1.40
671.90
18.9

OV_27.0RM
42.46
25.00
1.40
709.56
14.3

PASS:
WIDTH ON FACE:
HEIGHT:
PARTING:
AREA:
REDUCTION %:

9.0
9.61
9.12
1.00
65.33
16.8

9.5
10.15
9.64
1.00
73.00
14.0

10.0
10.67
10.14
1.20
80.75
19.2

10.5
11.20
10.64
1.20
88.91
17.6

11.0
11.74
11.16
1.20
97.82
18.4

11.5
12.28
11.66
1.20
106.84
15.8

12.0
12.80
12.16
1.20
116.13
14.6

12.5
13.34
12.68
1.20
126.28
19.0

13.0
13.88
13.18
1.20
136.43
18.0

13.5
14.39
13.68
1.20
146.98
19.8

14.0
14.95
14.20
1.20
158.37
17.8

14.5
15.46
14.70
1.20
169.72
14.4

15.0
15.90
15.22
1.40
181.94
21.0

15.5
16.53
15.72
1.40
194.09
19.2

16.0
17.07
16.22
1.40
206.63
21.2

17.0
18.13
17.24
1.40
233.43
17.6

18.0
18.0
19.23
1.40
261.87
16.3

19.0
19.0
20.31
1.40
291.34
21.6

20.0
20.0
21.41
1.40
323.02
20.1

21.0
22.51
21.28
1.38
355.66
18.3

22.0
23.58
22.30
1.40
390.57
21.6

23.0
24.68
23.32
1.40
427.12
20.0

24.0
25.78
24.34
1.40
465.30
20.6

25.0
26.86
25.34
1.40
504.32
20.4

26.0
27.95
26.36
1.40
545.73
18.7

27.0
27.0
27.38
1.40
588.79
17.0

COLD SIZE (MM)

9.009.5010.010.511.011.512.012.513.013.514.014.515.015.516.017.018.019.020.021.022.023.024.025.026.027.0

PASS:
WIDTH ON FACE:
HEIGHT:
PARTING:
AREA:
REDUCTION %:

PASS:
WIDTH ON FACE:
HEIGHT:
PARTING:
AREA:
REDUCTION %:

FINISHED SIZE:

9.009.5010.010.511.011.512.012.513.013.514.014.515.015.516.017.018.019.020.021.022.023.024.025.026.027.0

FINISHING M/S:
MTPH @ 100% EFF:
(PER STRAND)

42.475.037.975.034.375.031.175.028.375.025.975.023.875.021.975.020.375.018.875.017.575.016.375.015.275.014.375.013.475.011.975.010.675.09.5075.08.5775.07.7875.07.0975.06.4875.05.9575.05.4975.05.0775.04.7075.0

A BOX SELECTION:
B BOX SELECTION:
C BOX SELECTION:
D BOX SELECTION:

2222
2333
2444
--41--

NEW ROLL MOTOR RPM:

114710529308827466621243113910921028996899820798721685574483478456370354335312300267

REFERENCE DRAWINGS:

10289392 - ROLLING SCHEDULE STANDS AH - CV (BREAKDOWN MILL)
10289394 - ROLLING SCHEDULE FOR STANDS 1H TO 25
10289395 - DETAILS OF PASSES FOR STANDS 1H TO 25
10289396 - ROLLING SCHEDULE FOR THE RSM
10289396 - DETAILS OF PASSES FOR THE RSM

01ECN NO.

RevisionModification Contents

Date

Made by

Checked by

Condition

Sub Supplier

Similar To

Description 1

ROLLING SCHEDULE & PASS DETAILS

Date

Name

Description 2

SIZES: 9.0 TO 27.0MM

Made by

05/05/2016

KP

Description 3

REDUCING MILL ONLY

Checked by

MHG

Description 4

AM - GUJON

Approved by

KP

Description 5

Material No.

Version

Scale

NONE

Sheet

01

No. of Sheets

02

Dim. No.

10644111

Revision

01

DIMENSIONING ACCORDING TO ASME Y14.5M:1994
UNLESS OTHERWISE SPECIFIED:
ENVELOPE PRINCIPLE APPLIES.
ALL DIMENSIONS ARE IN MILLIMETERS.
BREAK ALL SHARP EDGES
COUNTERSINK ALL TAPPED HOLES TO MAJOR Ø
MACHINED SURFACE FINISH 6.3 (ASME Y14.36M)

REQUIREMENTS OF THE ABOVE APPLY

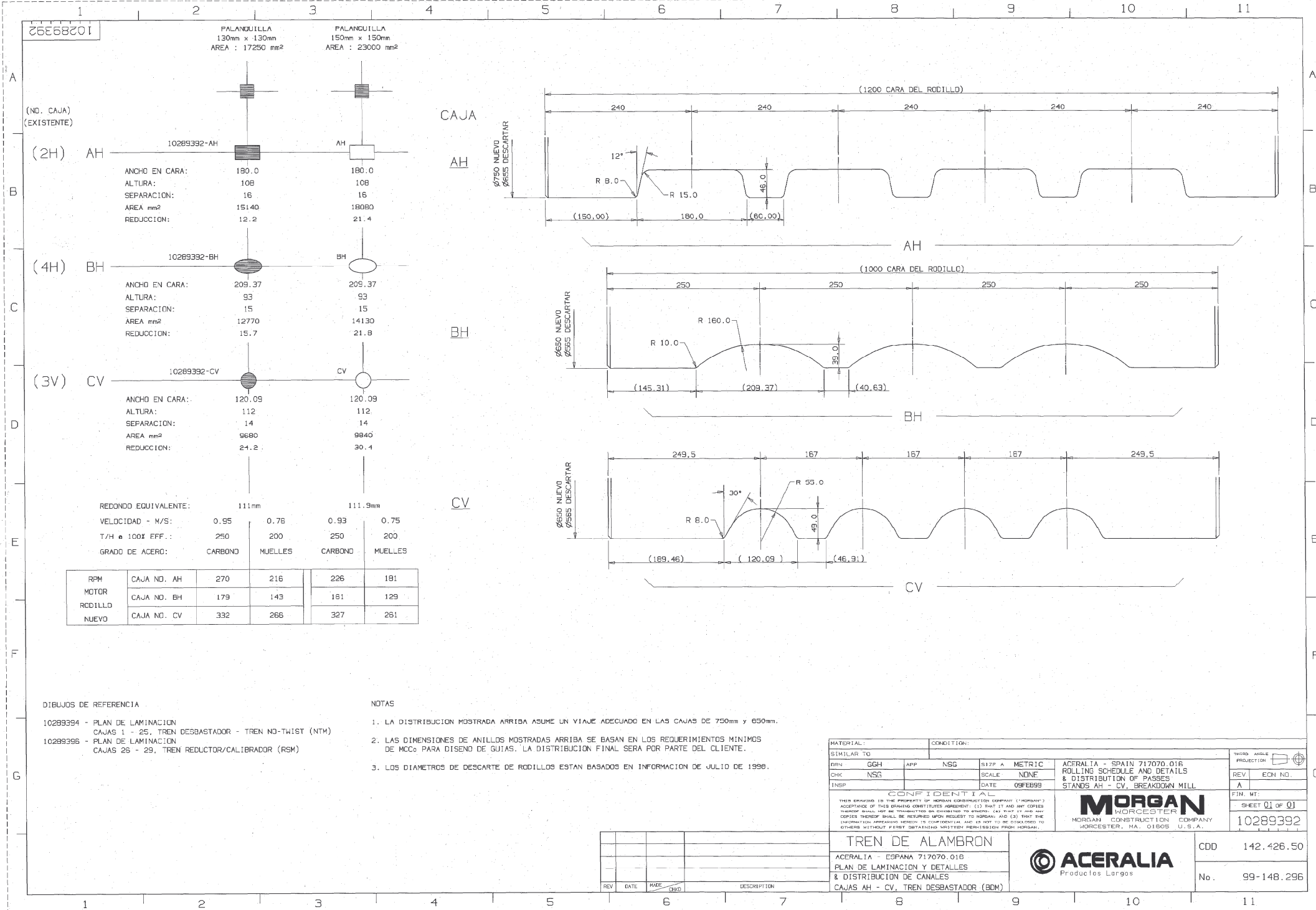
KP Thu May 05 11:29:53 2016

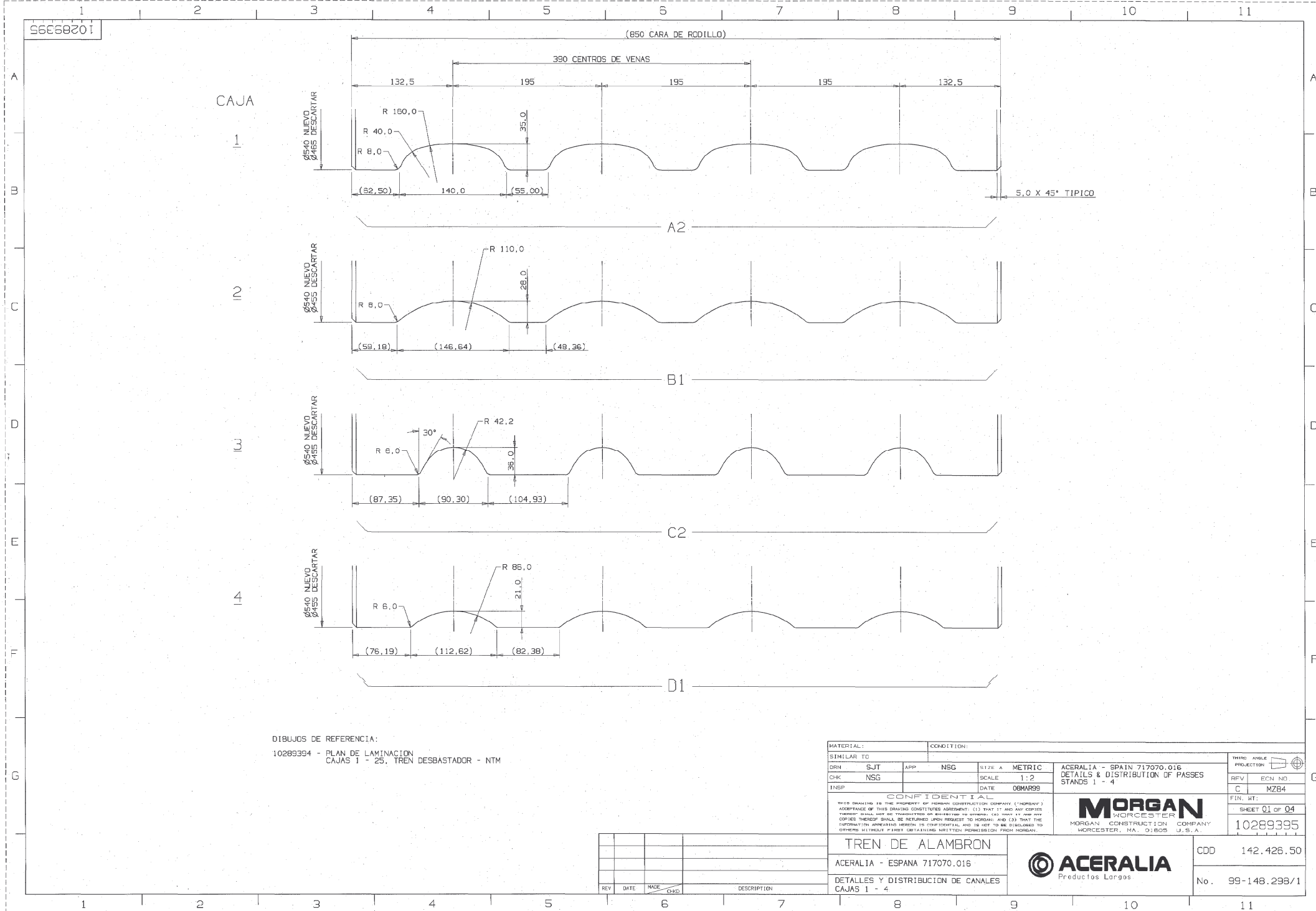
PRIMETALS TECHNOLOGIES

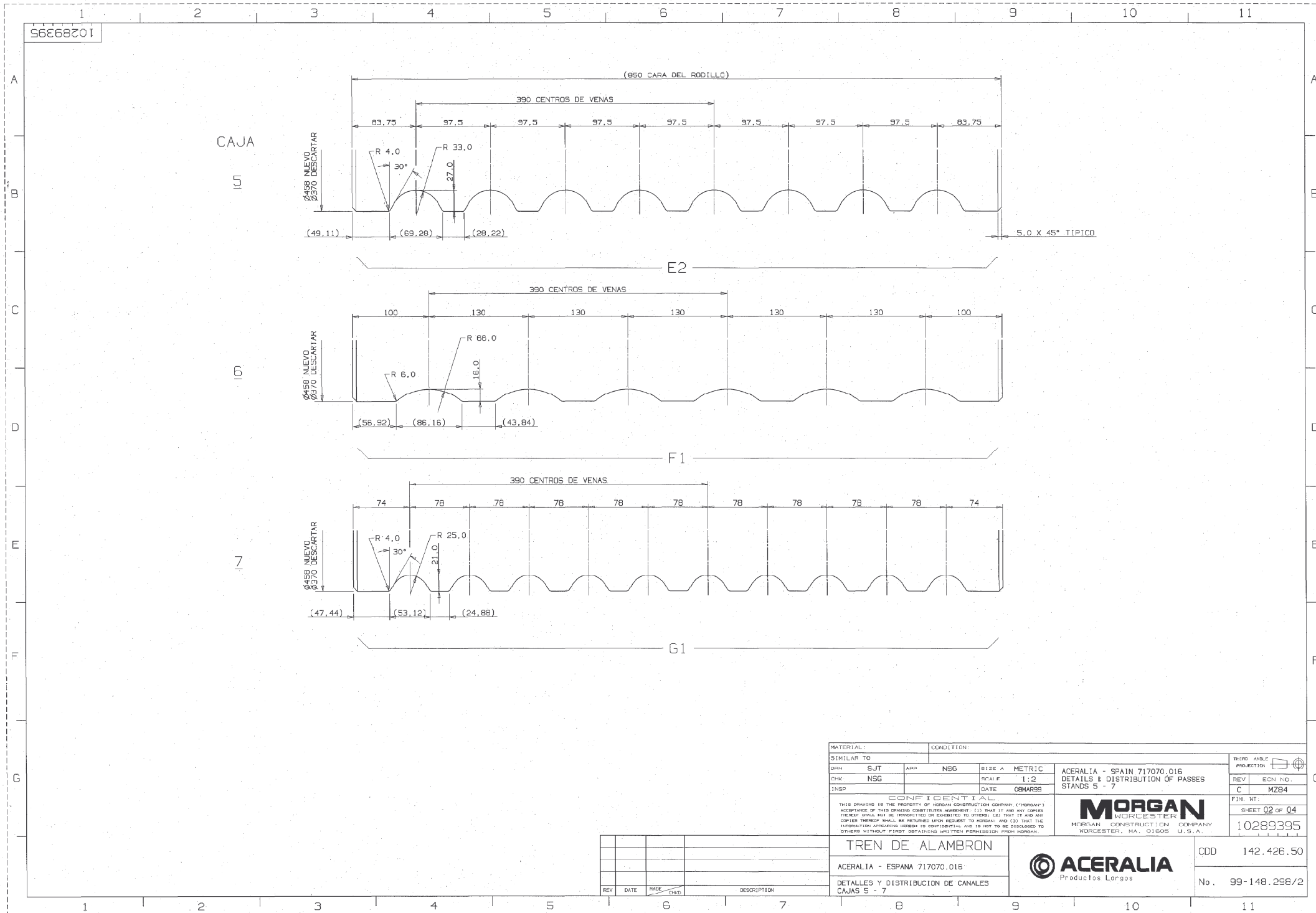
DIMENSIONING ACCORDING TO ASME Y14.5M:1994
UNLESS OTHERWISE SPECIFIED:
ENVELOPE PRINCIPLE APPLIES.
ALL DIMENSIONS ARE IN MILLIMETERS.
BREAK ALL SHARP EDGES
COUNTERSINK ALL TAPPED HOLES TO MAJOR Ø
MACHINED SURFACE FINISH 6.3/√(ASME Y14.36M)

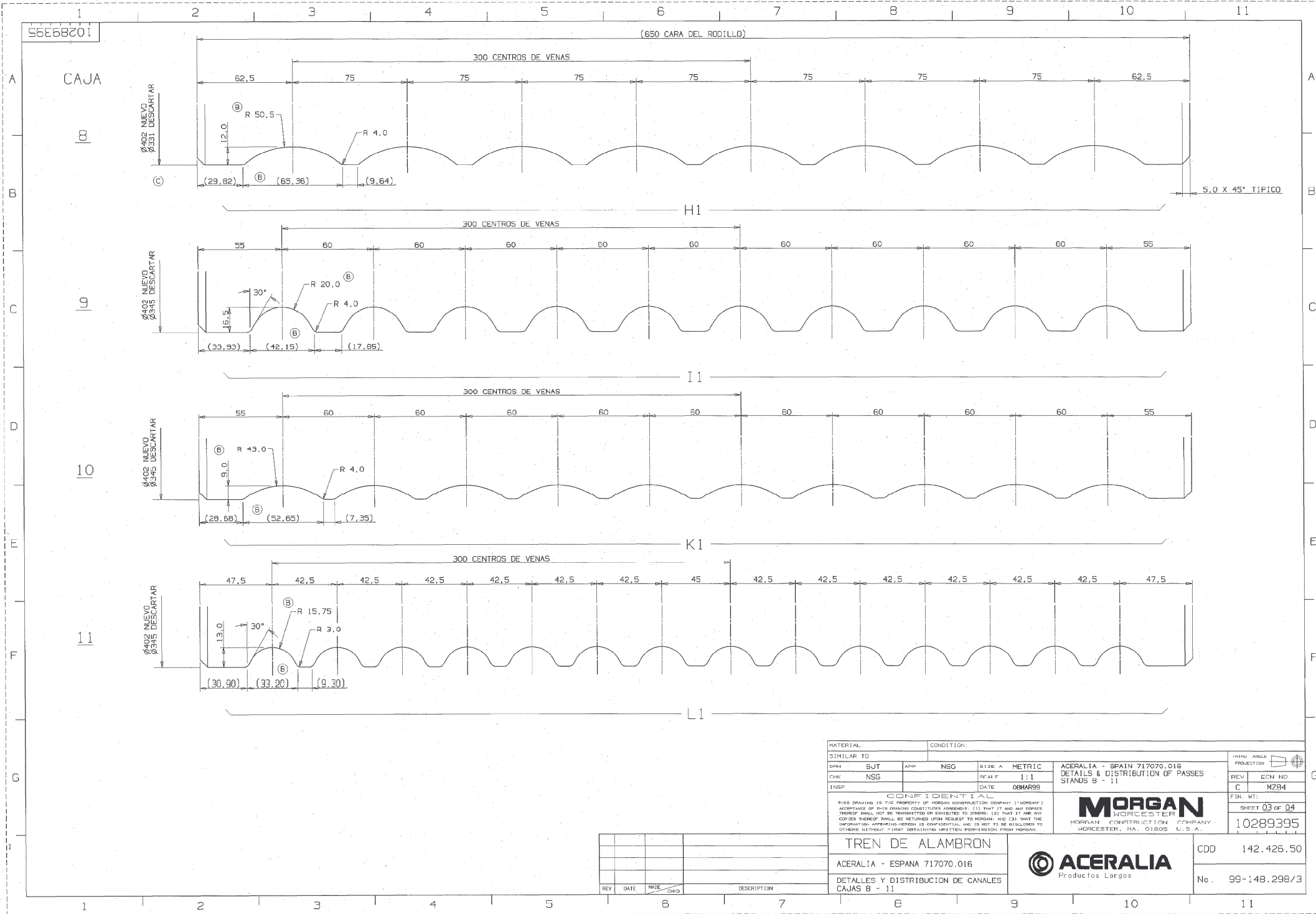
REQUIREMENTS OF THE ABOVE APPLY

01	ECN NO.		Date	Made by	Checked by
Revision	Modification Contents				
Condition		Sub Supplier	Similar to	Date	Name
Description 1		Made by	05/05/2016	KP	
Description 2		Checked by	MHG		
Description 3		Approved by	KP		
Description 4		Material	0.0	kg	
Description 5		Security Level			
Description 6		Status			
Description 7		Document Type			
Material No.		Version	Scale	NONE	Size A1
Dwg. No.		Sheet	01	No. of Sheets	02
Revision					01



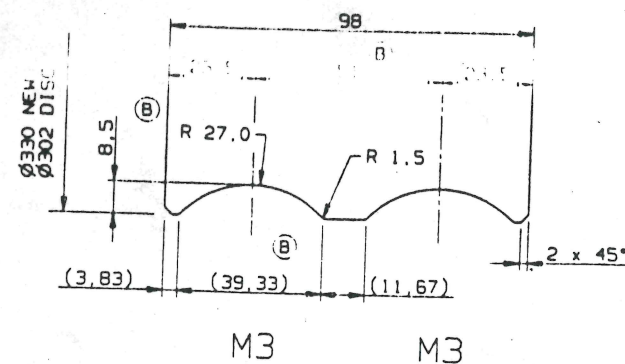




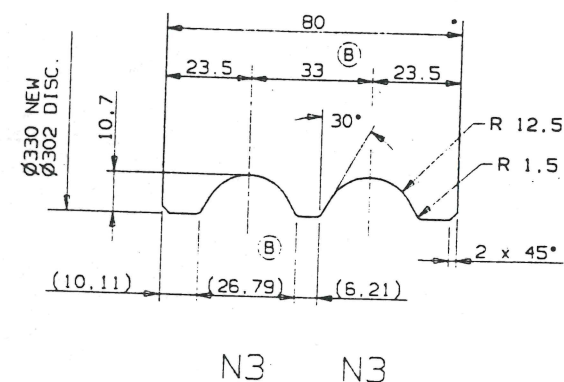


STAND

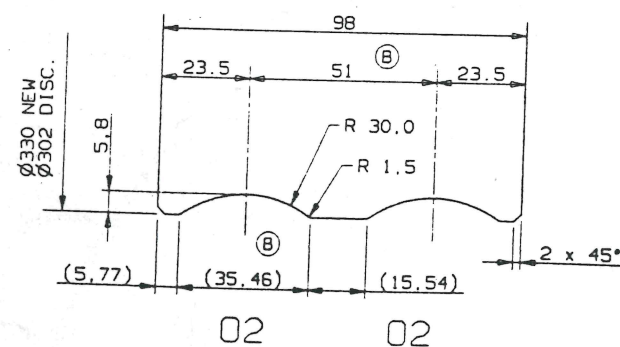
12



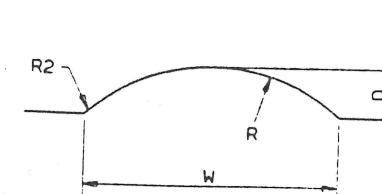
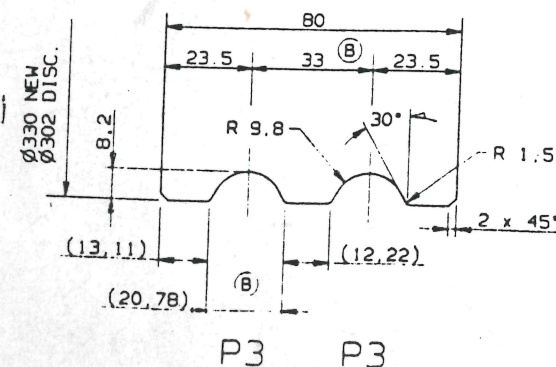
13V



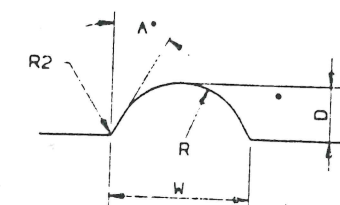
14



15V



TYPICAL OVAL GROOVES



TYPICAL ROUND GROOVES

NO-TWIST MILL STANDS NO. 16 - 25

STANDS 16 - 18 ROLL DIAMETER 210.50 NEW/189.50 DISC.
STANDS 19 - 25 ROLL DIAMETER 158.75 NEW/142.88 DISC.

100-75 NEW 142.06 DTSC

OVALS				
PASS	W (REF)	D (DEPTH)	R (GROOVE RADIUS)	R2 (RELIEF RADIUS)
R1	25.48	5.50	17.50	1.5
R5	27.95	5.50	20.50	1.5
T	18.89	3.50	14.50	1.5
T1	21.22	4.20	15.50	1.5
V	16.85	2.70	14.50	1.5
V1	17.66	3.00	14.50	1.5
X	13.56	2.10	12.00	1.5
X1	14.40	2.40	12.00	1.5
Z1	11.97	1.60	12.00	1.5

ROUNDS					
PASS	W (REF)	D (DEPTH)	R (GROOVE RADIUS)	R2 (RELIEF RADIUS)	A° (ANGLE)
S3	16.74	7.00	7.50	1.5	30
S4	17.90	7.00	8.50	1.5	30
14.0	14.24	6.10	7.15	--	15
12.5	12.90	5.50	6.35	--	20
11.0	11.30	4.75	5.60	--	20
10.0	10.24	4.25	5.10	--	20
8.5	8.78	3.50	4.33	--	25
Y	8.37	3.40	4.08	--	25
7.0	7.11	2.75	3.57	--	25
6.5	6.55	2.50	3.31	--	25

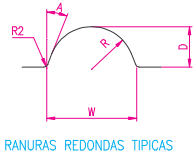
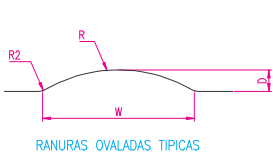
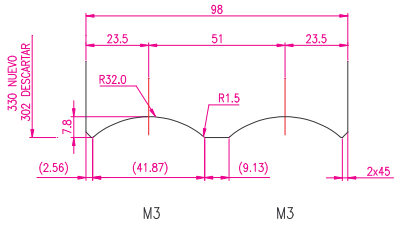
NOTE: ROLL DETAILS FOR STANDS 12 - 25 BY CUSTOMER. MCCo REFERENCE FOR STANDS 16 - 18, M199122A. FOR STANDS 19 - 25, M193784AA.

MATERIAL		CONDITION		THIRD ANGLE PROJECTION	
SIMILAR TO					
DRN	SJT	APP	NSG	SIZE A	METRIC
CHP	NSG			SCALE	1:1
DATE				08MAR99	
CONFIDENTIAL					
THIS DRAWING IS THE PROPERTY OF MORGAN CONSTRUCTION EQUIPMENT COMPANY. IT IS TO BE USED ONLY FOR THE PURPOSES INTENDED AND IS NOT TO BE REPRODUCED OR TRANSMITTED IN ANY FORM OR BY ANY MEANS, ELECTRONIC OR MECHANICAL, INCLUDING PHOTOCOPYING, RECORDING, OR BY ANY INFORMATION STORAGE AND RETRIEVAL SYSTEM. WITHOUT FIRST OBTAINING WRITTEN PERMISSION FROM MORGAN CONSTRUCTION EQUIPMENT COMPANY.					
ACERIALIA - SPAIN 717070.016 DETAILS & DISTRIBUTION OF PASSES STANDS 12 - 25				REV E B MF82 10289395	
MORGAN MORGAN CONSTRUCTION EQUIPMENT COMPANY WORCESTER, MA 01605 U.S.A.				AFT Thu May 13 09:53:18 1999	

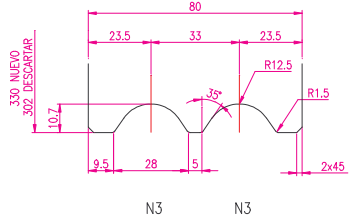
REV.	DESCRIPCION	NOMBRE	FECHA
A	Revisión general	Pruden	23/09/2005

CAJA

12

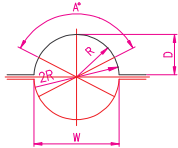
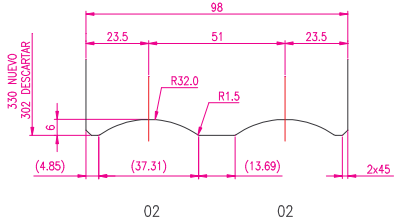


13V



TREN NO - TWIST CAJAS NO. 16 - 25					
CAJAS 16 - 18 RODILLO ø210,50 NUEVO/189,50 DESCARTAR					
CAJAS 19 - 25 RODILLO ø158,75 NUEVO/142,88 DESCARTAR					
OVALOS			REDONDOS		
CANAL	W (ANCHO DE CARA (REF.))	D (PROF.)	R (RADIO RANURA)	R2 (RADIO ALIVO)	A° (ANGULO)
R1	25.48	5.50	17.50	1.5	
R5	27.95	5.50	20.50	1.5	
T	18.89	3.50	14.50	1.5	
T1	21.22	4.20	15.50	1.5	
V	16.85	2.70	14.50	1.5	
V1	17.66	3.00	14.50	1.5	
X	13.56	2.10	12.00	1.5	
X1	14.40	2.40	12.00	1.5	
Z1	11.97	1.60	12.00	1.5	
S3	16.74	7.00	7.50	1.5	30
S4	17.90	7.00	8.50	1.5	30
14.0	14.24	6.10	7.15	---	15
12.5	12.90	5.50	6.35	---	20
11.0	11.30	4.75	5.60	---	20
10.0	10.24	4.25	5.10	---	20
8.5	8.78	3.50	4.33	---	25
Y	8.37	3.40	4.08	---	25
7.0	7.11	2.75	3.57	---	25
6.5	6.55	2.50	3.31	---	25

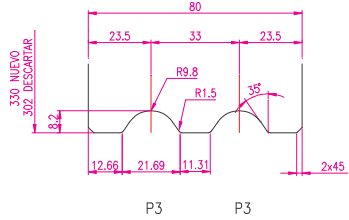
14



Redondos DOBLE radio						
Caja	CANAL	W (ANCHO DE CARA (REF.))	D (PROF.)	R (RADIO RANURA)	R2 (RADIO ALIVO)	A° (ANGULO)
19	FR14	14.72	6.15	7.10	1.5	117
21	FR11	11.65	4.85	5.58	1.5	114
23	FR9	9.42	3.80	4.56	1.5	113
25	PR6.5	7.69	3.10	3.68	1.5	109

REFERENCIA MORGAN: 10289395

15V



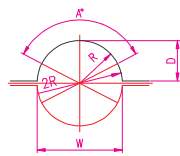
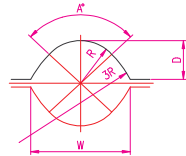
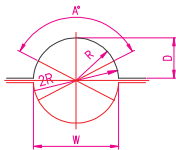
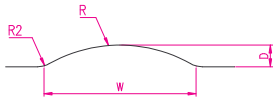
este plano sustituye al 99-148.298/4

Nº Piezas	DENOMINACION	Marcas	Materiales	Ubicación	Nº Plano	Pos
ACERALIA Corporación Siderúrgica, S.A.						
Responsable revisión	Nº DE PLANO	99-148.298/ H4 - A				Rev.
CODIFICACION		142/28.59				
FECHA	XX	INSTALACION	X			
DD / MM / AA	XX	EQUIPO	X			
Nº PROYECTO	XX	CONJUNTO	X			
	XX	SUBCONJUNTO	X			
	XX	PIEZA	X			
Fecha		Firma				
Dibujado 01/02		ABN				
Comprobado 01/02						
Aprobado						
Escala						
		DETALLES Y DISTRIBUCION DE CAÑALES CAJAS 12 - 25				
		Registro				Rev.
		01 1134 090				
						A1

NOTA: DETALLES DE RODILLOS PARA CAJAS 12 - 25 ver planos 99820055 y 99820056 PARA CAJAS 16 - 18, 99820043, PARA CAJAS 19 - 25, 99820044 y 99820045

REV.	DENOMINACION	NOMBRE	FECHA

REV.	DESCRIPCION	NOMBRE	FECHA
B	Revisión general	Pruden	23/09/2005



TREN REDUCTOR CAJAS 26 – 27									
DIMENSIONES DE CANAL CAJA 26					DIMENSIONES DE CANAL CAJA 27				
CANAL	W (ANCHO DE CARA (REF.))	D (PROF.)	R (RADIO RANURA)	R2 (RADIO ALMO)	CANAL	W (ANCHO DE CARA (REF.))	D (PROF.)	R (RADIO RANURA)	A° (ANGULO)
OV5A	9.48	1.35	9.00	1.5	PR5A	5.61	2.16	2.66	100
OV5B	10.16	1.31	10.50	1.5	PR5B	5.78	2.24	2.74	100
OV5.5	10.16	1.66	8.60	1.5	PR5.5	6.22	2.45	2.95	103
OV6	11.60	1.95	9.60	1.5	PR6	6.67	2.67	3.17	106
OV6.5	13.27	2.00	12.00	1.5	PR6.5	7.48	3.05	3.55	108
OV7	11.78	2.59	8.00	1.5	PR7	8.00	3.30	3.80	110
OV7.5	11.06	2.92	6.70	1.5	PR7.5	8.27	3.43	3.93	111
OV8	16.31	2.74	13.50	1.5	PR8	9.34	3.94	4.44	113
OV8.5	14.85	3.30	10.00	1.5	PR8.5	9.87	4.19	4.69	114
OV9-9.5	13.72	3.88	8.00	1.5	PR9-9.5A	10.40	4.44	4.94	115
					PR9.5B	11.21	4.72	5.32	113
OV10-10.5	19.68	3.68	15.00	1.5	PR10-10.5	11.74	4.98	5.58	114
OV11	18.27	4.22	12.00	1.5	PR11	12.28	5.23	5.83	114
OV11.5	17.03	4.76	10.00	1.5	PR11.5	12.80	5.48	6.08	115
OV12	24.37	4.42	19.00	1.5	PR12	13.88	5.99	6.59	116
OV12.5-13	22.64	5.16	15.00	1.5	PR12.5-13	14.67	6.37	6.97	117
OV13.5	21.95	5.64	13.50	1.5	PR13.5	15.22	6.62	7.22	117
OV14	21.44	5.89	12.70	1.5	PR14	15.46	6.75	7.35	118
OV14.5-16RB	27.61	5.90	19.10	1.5	PR14.5	17.07	7.41	8.11	117
OV15-16	26.96	6.34	17.50	1.5	PR15-16	17.62	7.67	8.37	117
OV16.5-17	25.66	7.23	15.00	1.5	PR16.5-17	18.67	8.17	8.87	118
OV17.5	24.59	7.93	13.50	1.5	PR17.5	19.50	8.55	9.25	118
OV18-18.5	33.08	7.77	21.50	1.5	PR18-18.5	21.09	9.31	10.01	119
OV19-19.5	31.68	8.43	19.10	1.5	PR19-19.5	21.86	9.69	10.39	120
OV20	31.54	8.83	18.50	1.5	PR20	22.41	9.95	10.65	120
OV20.5	30.89	9.27	17.50	1.5	PR20.5	22.95	10.20	10.90	120
OV21	30.50	9.48	17.00	1.5	PR21	23.22	10.33	11.03	120
OV21.5	29.59	9.91	16.00	1.5	PR21.5	23.77	10.58	11.28	120
OV22-23	39.54	9.70	25.00	1.5	PR22-23	25.34	11.34	12.04	121
OV23.5-24	37.57	10.55	22.00	1.5	PR23.5-24	26.43	11.85	12.55	121
OV24.5-25	36.69	11.35	20.50	1.5	PR24.5-25	27.51	12.36	13.06	121
OV25.5-26.5	37.65	12.38	20.50	1.5	PR25.5-26.5	29.35	13.24	13.94	122
OV6RB	10.24	1.60	9.00	1.5	FR6RB	6.41	2.54	3.04	104
OV8RB	15.76	2.33	14.50	1.5	FR8RB	8.54	3.56	4.06	111
OV10RB	19.15	3.06	16.50	1.5	FR10RB	10.67	4.47	5.07	112
OV12RB	28.12	9.00	30.00	1.5	FR12RB	12.80	5.48	6.08	115
OV16RB	27.61	5.90	19.10	1.5	FR16RB	17.07	7.41	8.11	117

* NOTA: ENTRE CAJAS 27 Y 29, LOS CANALES MARCADOS CON * SON COMUNES PARA AMBAS CAJAS.

Ⓡ ** NOTA: LOS CANALES DE CORRUGADO (RB) MOSTRADOS SON REDONDOS ACABADOS. CANALES DEFINITIVOS POR PARTE DEL CLIENTE.

DIBUJOS DE REFERENCIA:

99148299 – PLAN DE LAMINACION CAJAS 26 – 29, TREN REDUCTOR CALIBRADOR (RSM)
99930551 – RODILLO CSM DE 150 mm (DETALLE RODILLO 156mm x 44.45 ANCHO – TABLA – (1) CANAL)
99930550 – RODILLO CSM DE 150 mm (DETALLE RODILLO 156mm x 70 ANCHO – TABLA – (2) CANALES)
99930549 – RODILLO CSM DE 150 mm (DETALLE RODILLO 156mm x 57.275 ANCHO – TABLA – (4) CANALES)
99930548 – RODILLO H.H.D. DE 230mm (DETALLE RODILLO 228.34mm x 73.0 ANCHO – TABLA – (2) CANALES)
99930547 – RODILLO U.H.D. DE 230mm (DETALLE RODILLO 228.34mm x 57.275 ANCHO – TABLA – (4) CANALES)

NOTA: TOLERANCIAS DE CANAL EN TREN REDUCTOR/CALIBRADOR COMO SIGUEN:

CANAL	DEPTH	RADIUS
OVALO	+0.013 -0.013	+0.050 -0.000
REDONDO	+0.013 -0.013	+0.050 -0.000

NOTA: MATAR TODAS LAS ORILLAS FILOSAS

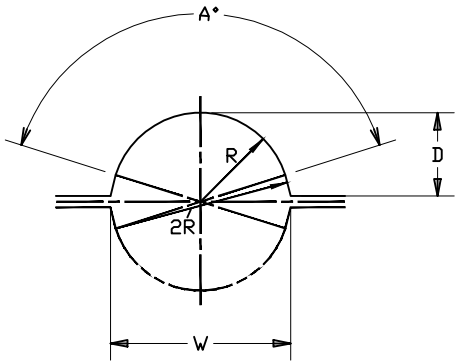
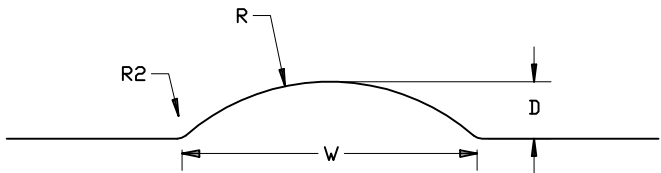
TREN CALIBRADOR CAJAS 28 – 29									
DIMENSIONES DE CANAL CAJA 28					DIMENSIONES DE CANAL CAJA 29				
CANAL	W (ANCHO DE CARA (REF.))	D (PROF.)	R (RADIO RANURA)	A° (ANGULO)	CANAL	W (ANCHO DE CARA (REF.))	D (PROF.)	R (RADIO RANURA)	A° (ANGULO)
LR5A	5.72	2.00	2.66	92	FR5A/B	5.26	2.04	2.54	105
LR5B	6.03	1.95	2.74	80	FR5.5	5.87	2.29	2.79	102
Ⓡ LR5.5	6.40	2.38	2.79	90	Ⓡ FR5.5	5.78	2.41	2.79	117
LR5.5	6.43	2.24	2.96	90	FR6	6.41	2.54	3.04	104
LR6	6.76	2.51	3.17	101	FR6.5	6.94	2.80	3.30	107
LR6.5	8.25	2.65	3.68	77	FR7	7.48	3.05	3.55	108
LR7	8.27	2.98	3.80	94	FR7.5	8.00	3.30	3.80	110
LR7.5	8.35	3.28	3.93	109	FR8	8.54	3.56	4.06	111
LR8	9.76	3.45	4.44	90	FR8.5	9.08	3.81	4.31	112
LR8.5	10.25	3.71	4.69	94	FR9	9.61	4.06	4.56	113
LR9	10.76	3.98	4.94	97	FR9.5A	10.15	4.32	4.82	114
LR9.5A	10.47	4.30	4.95	116	FR9.5B	10.15	4.22	4.82	111
Ⓡ LR9.5B	11.74	4.09	5.32	88	FR10	10.67	4.47	5.07	112
LR10	12.24	4.35	5.58	91	FR10.5	11.21	4.72	5.32	113
LR10.5	11.91	4.68	5.58	108	FR11	11.74	4.98	5.58	114
LR11	12.42	4.94	5.83	110	FR11.5	12.28	5.23	5.83	114
LR11.5	12.96	5.20	6.08	111	FR12	12.80	5.48	6.08	115
LR12	14.29	5.39	6.59	100	FR12.5	13.33	5.74	6.34	116
LR12.5	15.23	5.61	6.97	96	FR13	13.88	5.99	6.59	116
LR13	14.93	5.93	6.97	109	FR13.5	14.40	6.24	6.84	117
LR13.5	15.46	6.19	7.22	110	FR14	14.95	6.50	7.10	117
LR14	15.62	6.47	7.35	116	FR14.5	15.46	6.65	7.35	116
LR14.5	17.64	6.52	8.11	97	FR15	15.90	6.91	7.81	120
LR15	18.19	6.77	8.37	98	FR15.5	16.53	7.16	7.86	117
LR15.5	17.93	7.09	8.37	108	FR16	17.07	7.41	8.11	117
LR16	17.70	7.39	8.37	118	FR16.5	17.62	7.67	8.37	117
LR16.5	18.98	7.60	8.87	110	FR17	18.13	7.92	8.62	118
LR17	18.77	7.90	8.87	119	FR17.5	18.67	8.17	8.87	118
LR17.5	19.65	8.13	9.25	116	FR18	19.22	8.43	9.13	118
LR18	21.68	8.29	10.01	102	FR18.5	19.73	8.68	9.38	119
LR18.5	21.45	8.60	10.01	110	FR19	20.27	8.93	9.63	119
LR19	22.31	8.84	10.39	108	FR19.5	20.82	9.19	9.89	119
LR19.5	22.13	9.13	10.39	115	FR20	21.36	9.44	10.14	119
LR20	22.63	9.39	10.65	116	FR20.5	21.86	9.69	10.39	120
LR20.5	23.20	9.64	10.90	116	FR21	22.41	9.95	10.65	120
LR21	23.35	9.91	11.03	120	FR21.5	22.95	10.20	10.90	120
Ⓡ LR21.5	23.91	10.17	11.28	120	FR22	23.49	10.45	11.15	120
LR22	25.80	10.35	12.04	110	FR22.5	24.04	10.71	11.41	120
Ⓡ LR22.5	25.62	10.64	12.04	116	FR23	24.53	10.96	11.66	121
LR23	25.45	10.93	12.04	122	FR23.5	25.07	11.21	11.91	121
LR23.5	26.68	11.15	12.55	117	FR24	25.61	11.47	12.17	121
LR24	26.55	11.44	12.55	122	FR24.5	26.16	11.72	12.42	121
LR24.5	27.73	11.66	13.06	118	FR25	26.70	11.98	12.68	121
LR25	27.59	11.95	13.06	123	FR25.5	27.24	12.23	12.93	121
LR25.5	29.82	12.12	13.94	112	FR26	27.73	12.48	13.18	122
LR26	29.66	12.41	13.94	117	FR26.5	28.27	12.74	13.44	122
LR26.5	29.49	12.70	13.94	122					

este plano sustituye a 99930553

REFERENCIA MORGAN: 10289397

A	REVISION GENERAL	ABN	08	01	02
REV.	DENOMINACION	NOMBRE	FECHA		

		ACERALIA	
Compañía Siderúrgica, S.A.			
Responsable revisión	Nº DE PLANO	99.148.300 B	Rev.
CODIFICACION 142-08-01			
FECHA	XX	INSTALACION	-TREN DE ALAMBRON
DD / MM / AA	XX	EQUIPO	-CAJAS 26 A 28 DEL R. S. M.
Nº PROYECTO	XX	CONJUNTO	-PLAN DE LAMINACION
	XX	SUBCONJUNTO	
	XX	PIEZA	
Escalas	Fecha	Firma	
	12/01	A.C.V.	
Dibujado			
Comprobado		INGENIERIA, S.A.	
Aprobado		GUJON - ESPAÑA	
Escalas			
%	DETALLES DE CANALES CAJAS 26-29 TREN REDUCTOR/CALIBRADOR (RSM)		Registro
	01	1134	086
			A
			A1



REDUCING MILL STANDS 26 & 27

PASS DIMENSIONS STAND 26					NEW ROLL Ø 228.34 MM DISC. ROLL Ø 205.00 MM	PASS DIMENSIONS STAND 27				
PASS	W (WIDTH ON FACE) (REFERENCE)	D (DEPTH)	R (GROOVE RADIUS)	R2 (RELIEF RADIUS)		PASS	W (WIDTH ON FACE) (REFERENCE)	D (DEPTH)	R (GROOVE RADIUS)	A°(ANGLE)
OV_9.0RM	16.72	2.90	13.5	1.5		9.0	9.61	4.06	4.56	113
OV_9.5RM	15.22	3.51	10.0	1.5		9.5	10.15	4.32	4.82	114
OV_10.0RM	19.14	3.06	16.5	1.5		10.0	10.67	4.47	5.07	112
OV_10.5RM	19.19	3.47	15.0	1.5		10.5	11.20	4.72	5.32	113
OV_11.0RM	19.68	3.68	15.0	1.5		11.0	11.74	4.98	5.58	114
OV_11.5RM	18.24	4.22	12.0	1.5		11.5	12.28	5.23	5.83	114
OV_12.0RM	18.91	4.61	12.0	1.5		12.0	12.80	5.48	6.08	115
OV_12.5RM	22.35	4.36	16.5	1.5		12.5	13.34	5.74	6.34	116
OV_13.0RM	23.11	4.72	16.5	1.5		13.0	13.88	5.99	6.59	116
OV_13.5RM	25.16	4.76	19.0	1.5		13.5	14.39	6.24	6.84	117
OV_14.0RM	24.16	5.26	16.5	1.5		14.0	14.95	6.50	7.10	117
OV_14.5RM	21.44	5.89	12.7	1.5		14.5	15.46	6.75	7.35	118
OV_15.0RM	28.22	5.28	21.5	1.5		15.0	15.90	6.91	7.61	120
OV_15.5RM	27.25	5.76	19.0	1.5		15.5	16.53	7.16	7.86	117
OV_16.0RM	29.31	5.77	21.5	1.5		16.0	17.07	7.41	8.11	117
OV_17.0RM	27.27	6.53	17.5	1.5		17.0	18.13	7.92	8.62	118
OV_18.0RM	27.46	7.35	16.5	1.5		18.0	19.23	8.43	9.13	118
OV_19.0RM	32.66	7.26	22.0	1.5		19.0	20.31	8.93	9.63	118
OV_20.0RM	33.26	7.60	22.0	1.5		20.0	21.41	9.44	10.14	118
OV_21.0RM	34.51	8.35	22.0	1.5		21.0	22.51	9.95	10.65	118
OV_22.0RM	37.96	8.73	25.0	1.5		22.0	23.58	10.45	11.15	118
OV_23.0RM	39.13	9.44	25.0	1.5		23.0	24.68	10.96	11.66	118
OV_24.0RM	39.25	9.51	25.0	1.5		24.0	25.78	11.47	12.17	118
OV_25.0RM	40.56	10.38	25.0	1.5		25.0	26.86	11.97	12.67	118
OV_26.0RM	41.49	11.05	25.0	1.5		26.0	27.95	12.48	13.18	118
OV_27.0RM	42.46	11.80	25.0	1.5		27.0	27.96	12.99	13.69	140

NOTE: REDUCING/SIZING MILL PASS TOLERANCES AS FOLLOWS:		
PASS	DEPTH	RADIUS
OVAL	± 0.013 $- 0.013$	± 0.050 $- 0.000$
ROUND	± 0.013 $- 0.013$	± 0.050 $- 0.000$

- NOTE:
- BREAK ALL SHARP CORNERS
 - THE ROUND PASSES FOR 9.0 TO 26.0MM ARE SAME AS EXISITNG.

REFERENCE DRAWINGS:

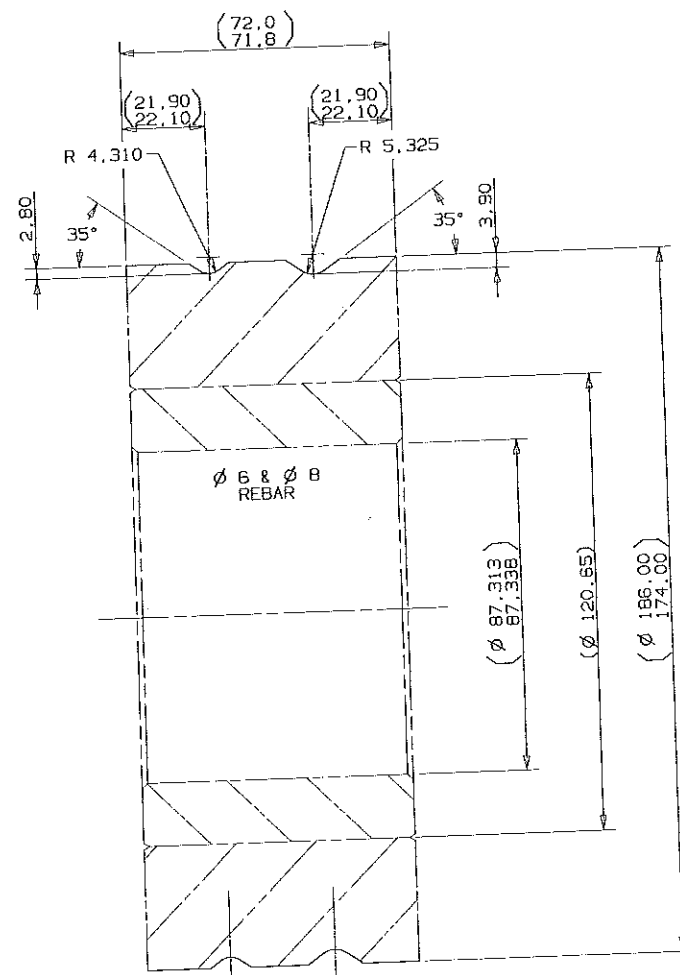
- 10289392 - ROLLING SCHEDULE STANDS AH - CV (BREAKDOWN MILL)
10289394 - ROLLING SCHEDULE FOR STANDS 1H TO 25
10289395 - DETAILS OF PASSES FOR STANDS 1H TO 25
10289396 - ROLLING SCHEDULE FOR THE RSM
10289396 - DETAILS OF PASSES FOR THE RSM

DIMENSIONING ACCORDING TO ASME Y14.5M:1994
UNLESS OTHERWISE SPECIFIED:
ENVELOPE PRINCIPLE APPLIES.
ALL DIMENSIONS ARE IN MILLIMETERS.
BREAK ALL SHARP EDGES
COUNTERSINK ALL TAPPED HOLES TO MAJOR Ø
MACHINED SURFACE FINISH $6.3\sqrt{}$ (ASME Y14.36M)

REQUIREMENTS OF
THE ABOVE APPLY

01	ECN NO.	Date	Made by	Checked by
Revision	Modification Contents			
Condition		Sub Supplier	Similar to	
Description 1		Made by	Date	Name
ROLLING SCHEDULE & PASS DETAILS		05/05/2016	KP	
Description 2		Checked by	MHG	
SIZES: 9.0 TO 27.0MM		Approved by	KP	
Description 3		Material		
REDUCING MILL ONLY		Name		
Description 4		Security Level		
Description 5		Status		
AM - GIJON		Document Type		
Material No.		Version	Scale	NONE
Draw. No.		Sheet	02	No. of Sheets
10644111		Revision	02	

10294315



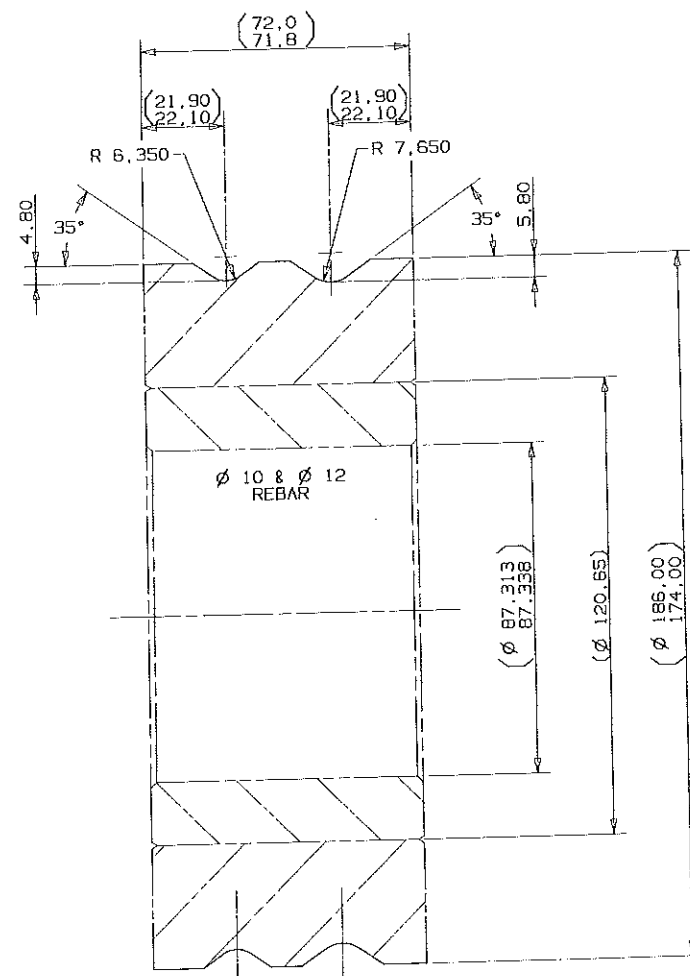
NOTES:

1. PASS GROOVES TO BE FINISHED 0.4 - 0.8 ROOT MEAN SQUARE.
2. BREAK SHARP EDGES.
3. AFTER GRINDING AND REGRINDING PASSES, THE OD AND PASSES OF PINCH ROLL SHOULD BE COATED. USE "RUST VETO A-2" CORROSION INHIBITOR TO PREVENT CORROSION IN THE NON-WORKING PASSES.
4. PASS GROOVE DISTRIBUTION IS A SUGGESTION ONLY. FINAL GROOVE DISTRIBUTION TO BE DETERMINED BY THE CUSTOMER.
5. ALL ROLL DIMENSIONS REFERENCED AND NOT SHOWN SHOULD BE HELD TO ORIGINAL ROLL DRAWING TOLERANCES AS LISTED ON 10290743 (7719460).

REFERENCE DRAWING:

10290743 (7719460) - PINCH ROLL (8" ROLL RING)

MATERIAL:		CONDITION:		THIRD ANGLE PROJECTION	
SIMILAR TO 10290935		PARTS MADE FROM THIS DWG MUST COMPLY WITH STD. MMS40073, MMS40075		REV	
DRN	TBT	APP	KP	SIZE A	METRIC
CHK	JFW			SCALE	1:1
INSP	CAR			DATE	08JUL1999
CONFIDENTIAL					
THIS DRAWING IS THE PROPERTY OF MORGAN CONSTRUCTION COMPANY ("MORGAN"). ACCEPTANCE OF THIS DRAWING CONSTITUTES AGREEMENT: (1) THAT IT AND ANY COPIES THEREOF SHALL NOT BE TRANSMITTED OR EXHIBITED TO OTHERS; (2) THAT IT AND ANY COPIES THEREOF SHALL BE RETURNED UPON REQUEST TO MORGAN; AND (3) THAT THE INFORMATION APPEARING HEREON IS CONFIDENTIAL AND IS NOT TO BE DISCLOSED TO OTHERS WITHOUT FIRST OBTAINING WRITTEN PERMISSION FROM MORGAN.					
MORGAN WORCESTER				FIN. WT: 26	
MORGAN CONSTRUCTION COMPANY				SHEET 01 OF 01	
WORCESTER, MA. 01605 U.S.A.				10294	
AFT Tue Jan 18 07:05:41 2000				11	



NOTES:

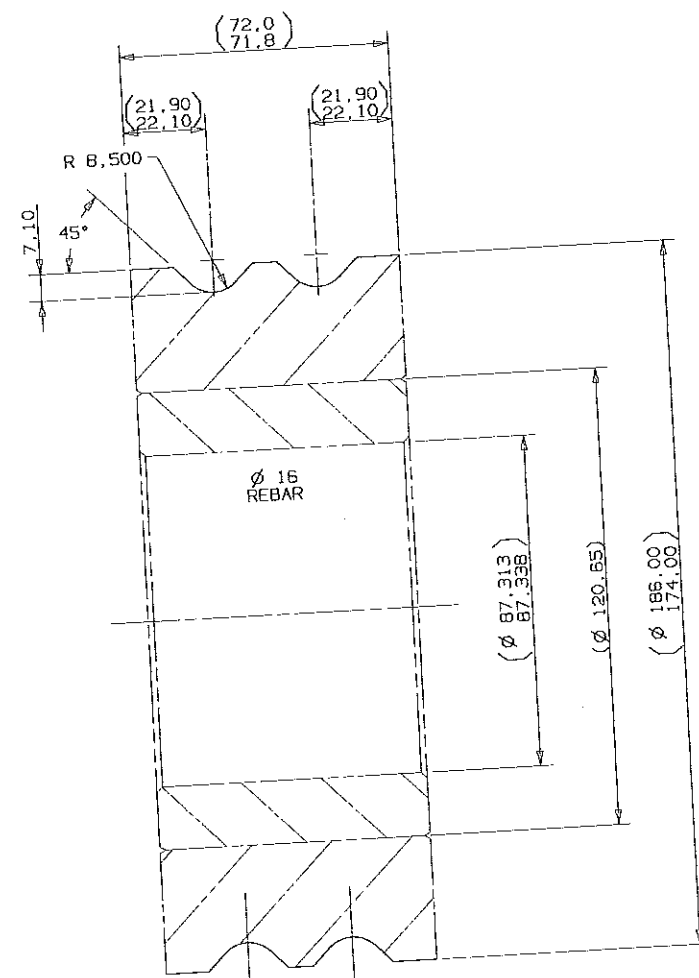
1. PASS GROOVES TO BE FINISHED 0.4 - 0.8 ROOT MEAN SQUARE.
2. BREAK SHARP EDGES.
3. AFTER GRINDING AND REGRINDING PASSES, THE OD AND PASSES OF PINCH ROLL SHOULD BE COATED. USE "RUST VETO A-2" CORROSION INHIBITOR TO PREVENT CORROSION IN THE NON-WORKING PASSES.
4. PASS GROOVE DISTRIBUTION IS A SUGGESTION ONLY. FINAL GROOVE DISTRIBUTION TO BE DETERMINED BY THE CUSTOMER.
5. ALL ROLL DIMENSIONS REFERENCED AND NOT SHOWN SHOULD BE HELD TO ORIGINAL ROLL DRAWING TOLERANCES AS LISTED ON 10290743 (7719460).

REFERENCE DRAWING:

10290743 (7719460) - PINCH ROLL (8" ROLL RING)

MATERIAL:		CONDITION:		THIRD ANGLE PROJECTION	
SIMILAR TO 10290936		PARTS MADE FROM THIS DWG MUST COMPLY WITH STD. MMS40073, MMS40075		REV	
DRN	TBT	APP	KP	SIZE A	METRIC
CHK	JFW			SCALE	1:1
INSP	CAR			DATE	06JUL1999
CONFIDENTIAL					
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MORGAN WORCESTER				FIN. WT: 26 k	
MORGAN CONSTRUCTION COMPANY				SHEET 01 OF	
WORCESTER, MA. 01605 U.S.A.				102943	
AFT Tue Jun 16 07:05:40 2000				11	

10294317



NOTES:

1. PASS GROOVES TO BE FINISHED 0.4 - 0.8 ROOT MEAN SQUARE.
2. BREAK SHARP EDGES.
3. AFTER GRINDING AND REGRINDING PASSES, THE OD AND PASSES OF PINCH ROLL SHOULD BE COATED. USE "RUST VETO A-2" CORROSION INHIBITOR TO PREVENT CORROSION IN THE NON-WORKING PASSES.
4. PASS GROOVE DISTRIBUTION IS A SUGGESTION ONLY. FINAL GROOVE DISTRIBUTION TO BE DETERMINED BY THE CUSTOMER.
5. ALL ROLL DIMENSIONS REFERENCED AND NOT SHOWN SHOULD BE HELD TO ORIGINAL ROLL DRAWING TOLERANCES AS LISTED ON 10290743 (7719460).

REFERENCE DRAWING:

10290743 (7719460) - PINCH ROLL (8" ROLL RING)

MATERIAL:		CONDITION:		THIRD ANGLE PROJECTION	
SIMILAR TO 10290937		PARTS MADE FROM THIS DWG MUST COMPLY WITH STD. MMS40073, MMS40075		REV	
ORN	TBT	APP	KP	SIZE A	METRIC
CHK	JFW			SCALE	1:1
INSP	CAR			DATE	08JUL1999
CONFIDENTIAL				ROLL, PINCH (DIA. 16 REBAR)	
THIS DRAWING IS THE PROPERTY OF MORGAN CONSTRUCTION COMPANY ("MORGAN"). ACCEPTANCE OF THIS DRAWING CONSTITUTES AGREEMENT: (1) THAT IT AND ANY COPIES THEREOF SHALL NOT BE TRANSMITTED OR EXHIBITED TO OTHERS; (2) THAT IT AND ANY COPIES THEREOF SHALL BE RETURNED UPON REQUEST TO MORGAN; AND (3) THAT THE INFORMATION APPEARING HEREON IS CONFIDENTIAL AND IS NOT TO BE DISCLOSED TO OTHERS WITHOUT FIRST OBTAINING WRITTEN PERMISSION FROM MORGAN.				MORGAN WORCESTER COMPANY WORCESTER, MA. 01605 U.S.A.	
				AFT Tue Jan 18 07:05:38 2000	

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11